Quantifying the Breakdown of Lambertian Photometric Stereo under Minnaert Reflectance

Nhi Ngo

Abstract

Photometric stereo recovers a per pixel surface normal from several images of a static scene under known, varying illumination, and the standard solver assumes the surface is Lambertian. Real surfaces are not. We quantify what that assumption costs. Writing the Minnaert reflectance model in its normalized form and viewing from a fixed viewpoint, the image stack reduces to a per pixel power law applied to Lambertian shading. The per pixel factor cancels in the recovered direction, so the estimated normal is unaffected by albedo, by source irradiance, and by the surface orientation relative to the camera, while the estimated albedo absorbs the full bias. We derive a first order expression for the angular error, confirm it against a controlled synthetic sweep with two analytically fixed endpoints, and evaluate the same solver on the DiLiGenT benchmark, reproducing the reference baseline to within 0.005 degrees on 9 of 10 objects. At the simplified Hapke point the error reaches 10.4 to 14.6 degrees across the light configurations we test, comparable to the mid tier objects of that benchmark. On the real objects every median effective exponent sits above one, a single global exponent underexplains the measured error, and an oracle per pixel exponent bounds what estimating the reduction could deliver at 9.02 degrees on average, below the best of the eight published classical methods we rescore here.

1. Introduction: shape from varying illumination

Recovering the three dimensional shape of a surface from images is a central problem in computer vision, and one of the oldest cues for it is shading. When a surface is lit from a known direction, its apparent brightness varies with the angle between the local surface orientation and the incident illumination, so brightness carries information about shape. Photometric stereo makes this practical by capturing several images of a static scene under different known light directions and solving for a surface normal at every pixel [15].

The technique matters wherever fine surface geometry has to be recovered without contact. Industrial inspection uses it to detect surface defects too shallow for range sensing, and image relighting uses the recovered normals to render a captured object under illumination it was never photographed in. Both applications share an assumption that is rarely examined: that the surface reflects light diffusely, in the Lambertian sense, so that observed brightness depends on the incident direction but not on the direction from which the surface is viewed.

Real materials violate this. The question this paper asks is not whether the assumption fails, which is well known, but by how much and as a function of what. We answer it by rendering surfaces whose reflectance is controlled by a single continuous parameter, running an unmodified Lambertian solver on all of them, and measuring the resulting angular error against ground truth that is exact by construction, then carrying the same machinery to real objects.

Our contributions are the following. First, we show that from a fixed viewpoint the Minnaert model collapses to a per pixel power law on Lambertian shading, that the per pixel factor cancels in the recovered direction, and that the same factor survives in full in the recovered albedo. Second, we derive a first order expression for the angular error, state the range over which it holds, and confirm both against a controlled sweep whose endpoints are fixed analytically. Third, we validate the solver on the DiLiGenT benchmark, matching the reference normal maps distributed with it to under 0.0003 degrees per object, and identify 73 degenerate ground truth pixels in one object that move its reported figure by 0.186 degrees depending on how they are scored. Fourth, we measure an effective exponent for every real object, show that a single exponent underexplains the measured error while an oracle per pixel exponent outperforms on average the eight published classical methods whose estimates ship with the benchmark, and show that the obvious way to estimate that exponent without ground truth fails.

2. Related work: non Lambertian photometric stereo

Photometric stereo originates with Woodham [15], who observed that three or more images under linearly independent lighting determine a Lambertian surface normal and albedo linearly at every pixel; the broader family of shading cues traces back to Horn [5]. The assumptions doing the work, a distant camera, distant calibrated lights, and diffuse reflectance, and the standard treatments of the linear solve appear in the textbook literature [14].

A long line of work removes the Lambertian assumption. Goldman et al. [3] fit a small set of parametric basis materials jointly with the normals. Alldrin et al. [1] make reflectance non parametric under isotropy. Robust formulations instead treat non Lambertian effects as outliers to reject: rank based recovery [16], consensus over illumination subsets [4], and sparse regression [8]. Model based alternatives include biquadratic reflectance [11], monotonicity constraints [12], and constrained bivariate regression [7]. The DiLiGenT benchmark [13] standardized evaluation for this line of work and distributes the estimated normal maps of the methods above, which we rescore under one metric in Table 2. Under that shared metric a decade of classical work improves on the least squares baseline by between 0.81 and 5.09 degrees on average. Learned approaches trained on synthetic reflectance libraries push further, to roughly seven or eight degrees on this benchmark [2, 6], and more recent designs go lower; nothing in this paper approaches that regime.

Our aim differs from all of the above. We do not propose a competitive method. We isolate a single scalar family of reflectance violations, derive what it does to the linear solver, and measure where the derivation stops holding. The correction that falls out of the derivation, raising each measurement to a power that linearizes the model, is elementary and is best read as the most degenerate special case of the parametric fitting line above [1, 3]: one exponent, no joint optimization. Its value here is diagnostic, an oracle bound on what the one parameter family can explain, not a claim of novelty.

3. Method: image formation and the linear solver

3.1. Lambertian image formation

Let $\hat{n}(x)$ be the unit surface normal at pixel x, $\rho(x)$ the albedo, and $\hat{s}_j$ the unit direction toward the jth of m known distant light sources. Under the Lambertian assumption, and away from shadow, the observed radiance is

$$I_j(x) \;=\; \rho(x)\,\big(\hat{n}(x)\cdot\hat{s}_j\big). \tag{1}$$

This is linear in the product $g = \rho\,\hat{n}$. Stacking the light directions as the rows of a matrix $L \in \mathbb{R}^{m\times 3}$ gives $I = Lg$ at every pixel, and for $m \geq 3$ light directions that are not coplanar in azimuth the system is overdetermined and well conditioned.

3.2. Recovering the normal

Minimizing the squared residual $\|I - Lg\|^2$ is an ordinary least squares problem whose normal equations give

$$g \;=\; (L^\top L)^{-1} L^\top I, \qquad \rho = \|g\|, \qquad \hat{n} = \frac{g}{\|g\|}. \tag{2}$$

The albedo is the length of the recovered vector and the normal is its direction. The solve is independent at each pixel and is applied to the whole image as a single matrix product.

3.3. Minnaert reflectance

To control how far the surface departs from Lambertian we use the Minnaert model [9] in its normalized form. With θ_i the incident angle and θ_r the viewing angle,

$$f_r \;=\; \frac{c+1}{2\pi}\,k(\lambda)\,\big(\cos\theta_i\cos\theta_r\big)^{c-1}, \qquad 0 \leq c \leq 1, \tag{3}$$

so that the outgoing radiance under a distant source of irradiance E_0 is

$$L_r \;=\; \frac{c+1}{2\pi}\,k(\lambda)\,E_0\,\cos^c\theta_i\,\cos^{c-1}\theta_r. \tag{4}$$

Setting $c = 1$ recovers the Lambertian case, and $c = 1/2$ gives the simplified Hapke model. We use the simplified Hapke form throughout. The full Hapke model, which additionally accounts for multiple scattering, shadowing and porosity, is not a special case of Minnaert, and no claim here extends to it.

3.4. Reduction to a power law

Fix the viewpoint. The viewing direction at a pixel then does not depend on which light is on, so $\cos\theta_r$ is a per pixel constant $\nu(x)$ across the image stack; under the orthographic viewing used in our experiments it is simply $n_z(x)$. Substituting into Equation (4), every view dependent factor collapses into one per pixel coefficient:

$$I_j(x) \;=\; A(x)\,\big(\hat{n}(x)\cdot\hat{s}_j\big)^c, \qquad A \;=\; \tfrac{c+1}{2\pi}\,k\,E_0\,\nu^{\,c-1}. \tag{5}$$

A Minnaert surface observed from a fixed viewpoint is exactly a per pixel power law applied to Lambertian shading. Two consequences follow immediately from Equation (2). Writing $w_j = (\hat{n}\cdot\hat{s}_j)^c$, the least squares solution is $g = A\,L^+ w$ with L^+ the pseudoinverse, so

$$\hat{n}_{\text{est}} = \frac{L^+ w}{\|L^+ w\|}, \qquad \rho_{\text{est}} = A\,\|L^+ w\|. \tag{6}$$

The positive scalar A cancels in the direction and survives in the length. The estimated normal therefore depends only on the exponent and the light directions, not on albedo, irradiance, or the orientation of the surface relative to the camera, while the estimated albedo absorbs the full $\nu^{\,c-1}$ bias, growing without bound toward the silhouette where $\nu \to 0$. Section 4.4 confirms both halves.

For the error itself, write $c = 1 - \varepsilon$ and expand $(\hat{n} \cdot \hat{s})^c$ to first order in ε:

$$\theta(x) \;\approx\; \varepsilon \left\|(\mathbb{I} - \hat{n}\hat{n}^\top)\, L^+ u\right\|, \qquad u_j = (\hat{n} \cdot \hat{s}_j)\ln(\hat{n} \cdot \hat{s}_j), \tag{7}$$

where θ is the angular error in radians. The coefficient A cancels here as well, which independently confirms the invariance above. At the far end of the range the model makes a second exact prediction: at $c = 0$ every unshadowed observation equals A, so $g \propto L^+\mathbf{1}$ and the estimated normal field collapses to one direction at every fully lit pixel, regardless of the true surface. The mean error at $c = 0$ is therefore computable in closed form before running anything.

Both predictions hold on pixels seen by every light. Where $\hat{n} \cdot \hat{s}_j \leq 0$ the observation clamps to zero rather than following the power law, which is why shadow handling appears as a separate ablation in Section 4.5.

It is worth being precise about what the derivation actually assumes, because it is weaker than it first appears. Nothing above requires orthographic projection. All that is needed is a viewpoint that does not move across the light stack, so that ν carries no light index whatever its value at each pixel. A finite camera therefore changes ν from a constant into a field but leaves Equation (5) intact. Substituting such a field into the renderer confirms the algebra numerically: individual radiances move by up to 43 percent while the recovered normals move by 8e-15 degrees, which is machine precision. The orthographic capture of the benchmark is convenient, not load bearing.

4. Experiments: synthetic and real

4.1. Benchmark and protocol

We evaluate on the DiLiGenT photometric stereo benchmark, 10 real objects spanning a wide range of reflectance, from a near diffuse ball to strongly specular metallic pieces [13]. Each object provides 96 images at 612×512 under calibrated directional lighting, together with a foreground mask and a ground truth normal map obtained independently of the photometric measurements. The capture is orthographic and single view, which satisfies the fixed viewpoint assumption of Section 3.4.

Each observation is divided by the measured intensity of its light source before the colour channels are combined, following the protocol distributed with the benchmark. The solver of Equation (2) is applied without modification, that is, with no changes made for real input relative to the synthetic experiments. Synthetic experiments use a unit sphere with analytic normals, so their ground truth carries no discretization error, and a smooth height field as a second geometry.

Object	Ours	Reference	Δ
ball	4.10	4.10	-0.004
bear	8.39	8.39	-0.001
buddha	14.92	14.92	+0.001
cat	8.41	8.41	+0.003
cow	25.60	25.60	-0.001
goblet	18.50	18.50	-0.000
harvest	30.62	30.62	+0.005
pot1	8.89	8.89	+0.004
pot2	14.49	14.65	-0.156
reading	19.80	19.80	+0.003
Average	15.37	15.39	-0.017

Table 1. Mean angular error in degrees on the benchmark, against the reference Lambertian baseline distributed with it. 9 of 10 objects agree to within 0.005 degrees. The exception, pot2, is discussed in Section 4.3.

4.2. Evaluation

We report the mean angular error in degrees between the estimated and ground truth normals over the foreground mask,

$$\varepsilon(x) \;=\; \operatorname{atan2}(\|\hat{n}_{\text{est}} \times \hat{n}_{\text{gt}}\|,\; \hat{n}_{\text{est}} \cdot \hat{n}_{\text{gt}})\,. \tag{8}$$

The two argument form is used rather than the arc cosine of the dot product because the derivative of $\arccos$ is unbounded at zero angle, which limits its usable resolution to roughly 10^{-6} degrees. That matters here because the synthetic Lambertian case is exact to machine precision and would otherwise be reported at the resolution limit of the metric rather than of the solver. Pixels whose ground truth normal has zero length are excluded rather than scored, since the angle to a vector with no direction is undefined; the consequences of this choice are quantified in Section 4.3.

4.3. Baseline results on real objects

Table 1 reports the result on all 10 objects. The average of 15.37 degrees agrees with the reference baseline, and 9 of the 10 objects match to within 0.005 degrees. Beyond the aggregate, the estimated normal maps agree with the reference normal maps distributed with the benchmark to under 0.0003 degrees per object, which fixes the solver and the data handling conventions pixel by pixel rather than only in summary.

Two details of the protocol are worth recording because they change the reported figures. The reference pipeline converts colour to grey with a routine that clamps its output to the unit interval. Without that clamp, saturated pixels inside specular highlights disagree by up to 16 degrees and the ball error reads 4.17 rather than 4.10. The clamp affects only 0.02 percent of observations, but those observations concentrate in highlights and lower the reported error, so it

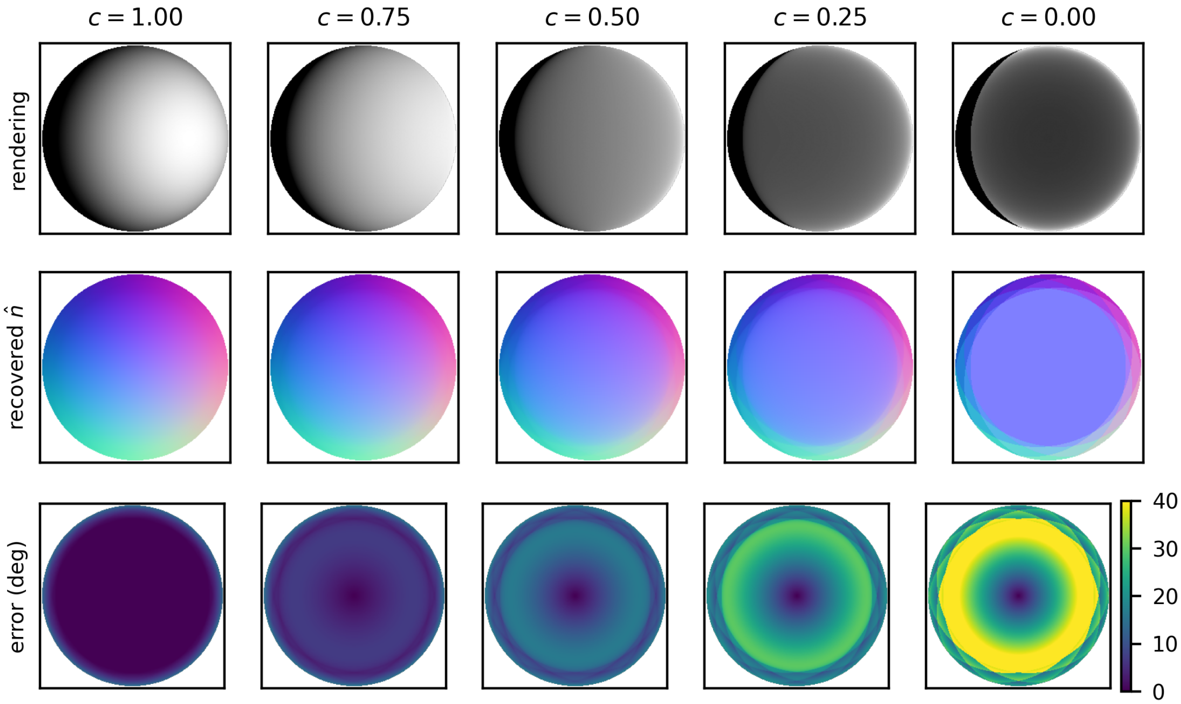

Figure 1. The breakdown, end to end. A sphere rendered under one of 10 lights at five Minnaert exponents (top), the normal map recovered by the unmodified Lambertian solver (middle), and its angular error (bottom). At $c = 1$ recovery is exact to machine precision. As c falls the recovered field flattens toward a single direction, and at $c = 0$ it collapses to exactly that: the uniform color predicted in closed form by Section 3.4.

acts as a mild outlier suppressor inside what is presented as an unmodified Lambertian baseline.

The second detail concerns pot2. Its ground truth normal map contains 73 pixels of exactly zero length inside the mask, so the true orientation there is undefined. A metric that evaluates $\arccos$ of the dot product scores those pixels at 90 degrees, while a magnitude invariant formulation silently scores them at zero, that is, as perfect matches for any estimate whatsoever. Those two conventions differ by 0.186 degrees on this object. We adopt neither, excluding the pixels and reporting the count, which leaves the 0.156 degree gap visible in Table 1. That the accounting is exact can be checked in one line: scoring the excluded pixels at 90 degrees and averaging over the full mask reproduces the tabulated 14.65 to four decimal places.

4.4. The controlled breakdown

Figure 1 shows the experiment end to end and Figure 2 quantifies it. A sphere is rendered at 51 exponents from $c = 1$ down to $c = 0$ and the unmodified solver runs on each stack; error is measured on fully lit pixels. Three facts anchor the curve. At $c = 1$ the mean error is 7.2e-15 degrees, exact recovery at machine precision. At $c = 0$ the measured 34.30 degrees matches the closed form collapse

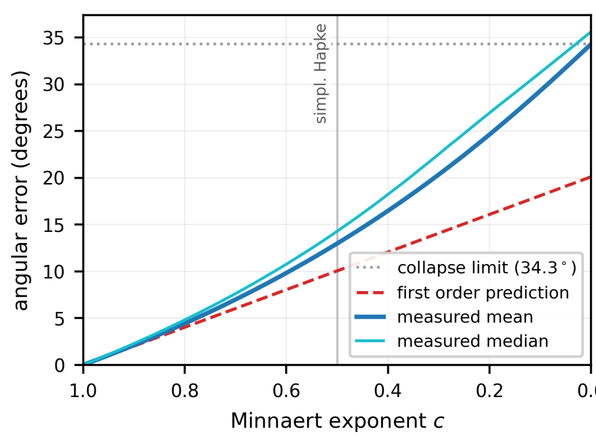

Figure 2. Angular error of the Lambertian solver on a Minnaert sphere, 10 lights. Both endpoints are fixed by theory: machine precision at $c = 1$ and the closed form collapse limit at $c = 0$, which the measurement meets to two decimals. The first order prediction of Equation (7) tracks the measurement near $c = 1$ and understates it below, by $1.29\times$ at the simplified Hapke point and $1.71\times$ at $c = 0$.

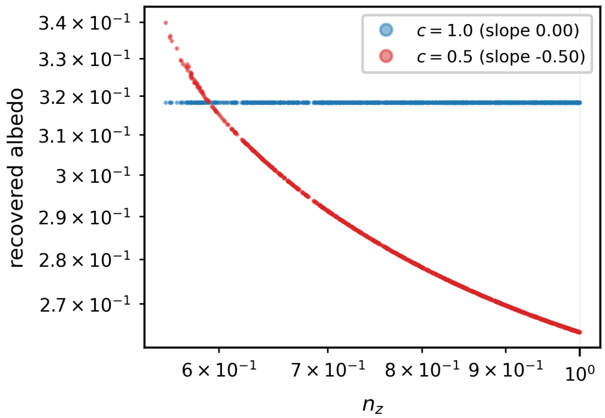

Figure 3. The other half of the invariance. Recovered albedo against n_z on fully lit pixels. At $c = 1$ the albedo is exact and flat. At $c = 0.5$ it follows the predicted n_z^{c-1} power law, isolated slope -0.50, while the recovered normals in both cases are the same to within 10^{-10} degrees. The reflectance violation is absorbed almost entirely by the albedo channel.

limit of 34.30 degrees, and the recovered field is a single direction, visible as the uniform color in Figure 1. In between the error is strictly monotone, reaching 12.98 degrees at the simplified Hapke point under this rig.

Two qualifications keep that headline honest. The magnitude is rig dependent: across 12 light configurations of varying count and slant the simplified Hapke error spans 10.4 to 14.6 degrees, so any single figure must carry its rig. And the first order law has a stated range: it tracks the measurement within ten percent for $c \geq 0.9$ and degrades into a lower bound beyond, understating the true error by 1.29× at $c = 0.5$. The prediction is useful where perturbation theory says it should be, and visibly labeled where it is not. The curve itself is invariant to the albedo pattern and to image scale, as Equation (6) requires, and repeating the sweep on the height field geometry preserves its shape exactly, exact at $c = 1$ and monotone below, while the magnitude drops to 7.4 degrees at the simplified Hapke point because that surface presents a narrower range of orientations: median tilt 13 degrees against the sphere's 36.

The split predicted by Equation (6) is shown directly in Figure 3. At $c = 0.5$ the recovered albedo follows the n_z^{c-1} bias, isolated slope -0.50 against a predicted -0.5, and diverges toward the silhouette, while the normals it accompanies are bitwise identical to the ones recovered with the bias removed. One further distributional note: across the whole sweep the median error sits above the mean, because pixels whose normals align with the rig axis are biased least and pull the mean down.

4.5. Ablations

Figure 4 isolates the other levers. Conditioning behaves as the classical analysis says it should: squeezing the light azimuths into a narrow wedge drives $\mathrm{cond}(L)$ from 2.02 to 1335 and the error from about one degree to 74.3, monotonically and with log log correlation 0.992, though sublinearly rather than in proportion, and the most degenerate rigs saturate at errors no better than pointing every normal at the camera. With well spread lights, error falls with light count at a fitted exponent of -0.50, the $1/\sqrt{m}$ averaging rate, and grows linearly in noise, fitted exponent 1.00.

Shadow handling is the ablation with a twist. Discarding observations below the local horizon makes Lambertian recovery exact on the full sphere, removing the only error source. On a Minnaert surface the same masking helps only down to $c \approx 0.70$ and then hurts, because the reflectance bias is systematic rather than an outlier: retaining the clamped observations happens to shrink the estimate in a direction that partially cancels the bias, and below the crossover that accident is worth more than the bad rows cost. Robustness machinery tuned for outliers does not automatically transfer to model mismatch.

4.6. Real materials and the effective exponent

The reduction can be run in reverse. Taking logarithms of Equation (5), $\log I_j = \log A + c \log(\hat{n} \cdot \hat{s}_j)$ is linear per pixel, so with the benchmark's ground truth normals the slope of a per pixel least squares line is an effective exponent c_{eff}: a measurement of how far each material departs from Lambertian, in the only parameter the model has. On synthetic Minnaert data the fit recovers the true exponent to 10^{-8}. On the real objects it produces the medians in Table 3, and they are the first surprise: every object has a median $c_{\text{eff}} > 1$, from 1.08 to 2.14, outside the Minnaert domain entirely. Real glossy materials brighten faster than the cosine, they are super Lambertian, where the Minnaert family with $c < 1$ describes the opposite, lunar like flattening. The claim is about medians rather than every pixel: on 6 of the 10 objects the lower quartile still falls below one, and on the most specular object the upper quartile sits at the limit of the range the fit is permitted to report, so its median understates the departure.

Figure 6 then asks whether that one number explains the measured error, by rendering the sphere at each object's c_{eff} under that object's own light rig, so that the comparison is not confounded by rig dependence. It does not: every object sits above the diagonal, by a factor of 1.7 for the most diffuse objects up to 7.7 for the most specular. A single global exponent is the wrong model for real materials, and Figure 5 shows why directly. The exponent is not a property of an object but a field across it: nearly uniform on the near diffuse ball, spiking on the cat's glossy patches, and saturating across harvest's cloth while its figures stay low. In each

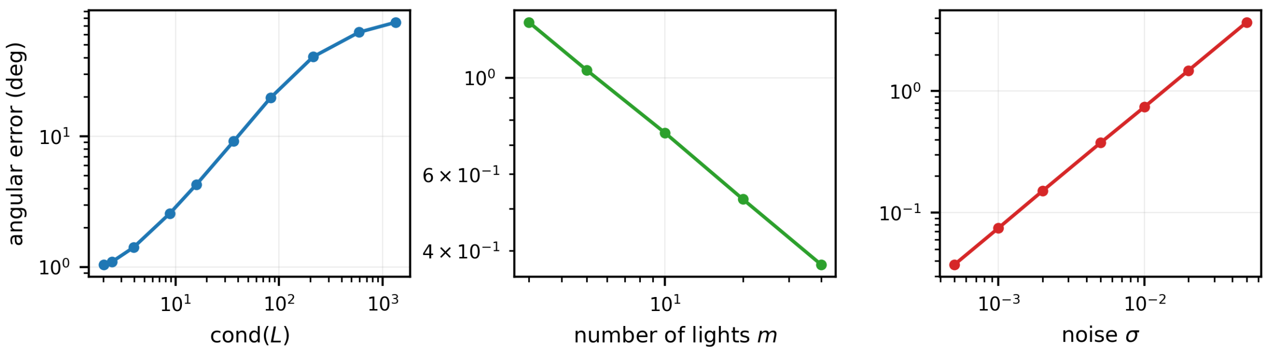

Figure 4. What else moves the error, under noise $\sigma = 0.01$ with an independent draw per level. Left: error against the condition number of the light matrix as azimuths collapse toward a common value; growth is monotone and strong (log log correlation 0.992) but sublinear, and the most degenerate rigs saturate. Center: error against light count, fitted exponent -0.50, the averaging rate. Right: error against noise level, fitted exponent 1.00 in the small noise regime.

Method	ball	bear	buddha	cat	cow	goblet	harvest	pot1	pot2	reading	avg
Shi et al. [11]	1.74	6.12	10.60	6.12	13.93	10.09	25.44	6.51	8.61	13.63	10.28
Ikehata et al. [7]	3.34	7.11	10.47	6.74	13.05	9.71	25.95	6.64	8.60	14.19	10.58
Goldman et al. [3]	1.85	5.62	13.67	7.27	8.46	11.56	27.13	7.52	6.61	18.02	10.77
Higo et al. [4]	3.55	10.87	12.00	7.90	14.39	12.57	20.88	10.43	15.58	16.03	12.42
Alldrin et al. [1]	2.71	5.96	12.54	6.53	21.48	13.93	30.50	7.23	10.87	14.17	12.59
Wu et al. [16]	2.06	6.50	10.91	6.73	25.89	15.70	30.01	7.18	12.96	15.39	13.33
Ikehata et al. [8]	2.54	7.14	11.11	7.11	25.58	16.25	29.17	7.65	13.91	16.17	13.66
Shi et al. [12]	13.58	19.44	18.37	12.34	7.62	17.80	19.30	10.37	9.67	17.17	14.57
Least squares baseline [15]	4.10	8.39	14.92	8.41	25.60	18.50	30.62	8.89	14.49	19.80	15.37

Table 2. Reproduced evaluation of the estimate maps distributed with the benchmark [13], scored with the metric of Section 4.2: mean angular error in degrees, degenerate ground truth pixels excluded. These are published results rescored for comparability, not results of this paper. The pot2 column sits slightly below originally published values for every method because of the exclusion discussed in Section 4.3.

case the exponent map and the error map beside it share the same spatial structure, which is exactly the information a per object median discards.

The per pixel version of the same idea is where the signal lives. Equation (5) implies its own correction: raising each observation to $1/c$ makes the stack exactly Lambertian, and with the true exponent this restores machine precision recovery on synthetic data, at every $c > 0$, by identity. Applying it on real data with the per pixel oracle fit gives the Oracle column of Table 3: 9.02 degrees on average, an improvement on every single object, and below the best of the eight rescored classical methods in Table 2 (10.28 degrees). A per object exponent manages only 13.82. The one parameter family, allowed to vary spatially, explains more of the real error than a decade of published classical machinery recovered; the caveat, and it is the whole caveat, is that the oracle reads the answer key.

Two legal comparisons calibrate that caveat. Estimating the exponent without ground truth by alternation, fitting against the current normal estimate and resolving, fails outright: three rounds end at 18.30 degrees, worse than the uncorrected baseline, because an exponent fitted against biased normals reinforces their bias. And trimmed least squares, which drops the darkest and brightest quarter of observations at each pixel and uses no reflectance model at all, reaches 12.03 degrees. The information the oracle exploits is demonstrably present in the data, but the obvious estimator of it destroys it, and a model free robust baseline captures a substantial fraction of the available gain for far less machinery.

5. Discussion and limitations

The synthetic finding is compact: under a fixed viewpoint, one scalar reflectance parameter maps to angular error through a curve with analytically pinned endpoints, first order behavior near Lambertian, and complete information loss at $c = 0$, while the albedo channel absorbs the entire per pixel bias that the normal channel provably cannot see. Moderate reflectance violations are already expensive: the simplified Hapke point costs 10.4 to 14.6 degrees, an error comparable to mid tier real objects, and this is a floor imposed by the reflectance model alone, before noise, shadows, or calibration error are added.

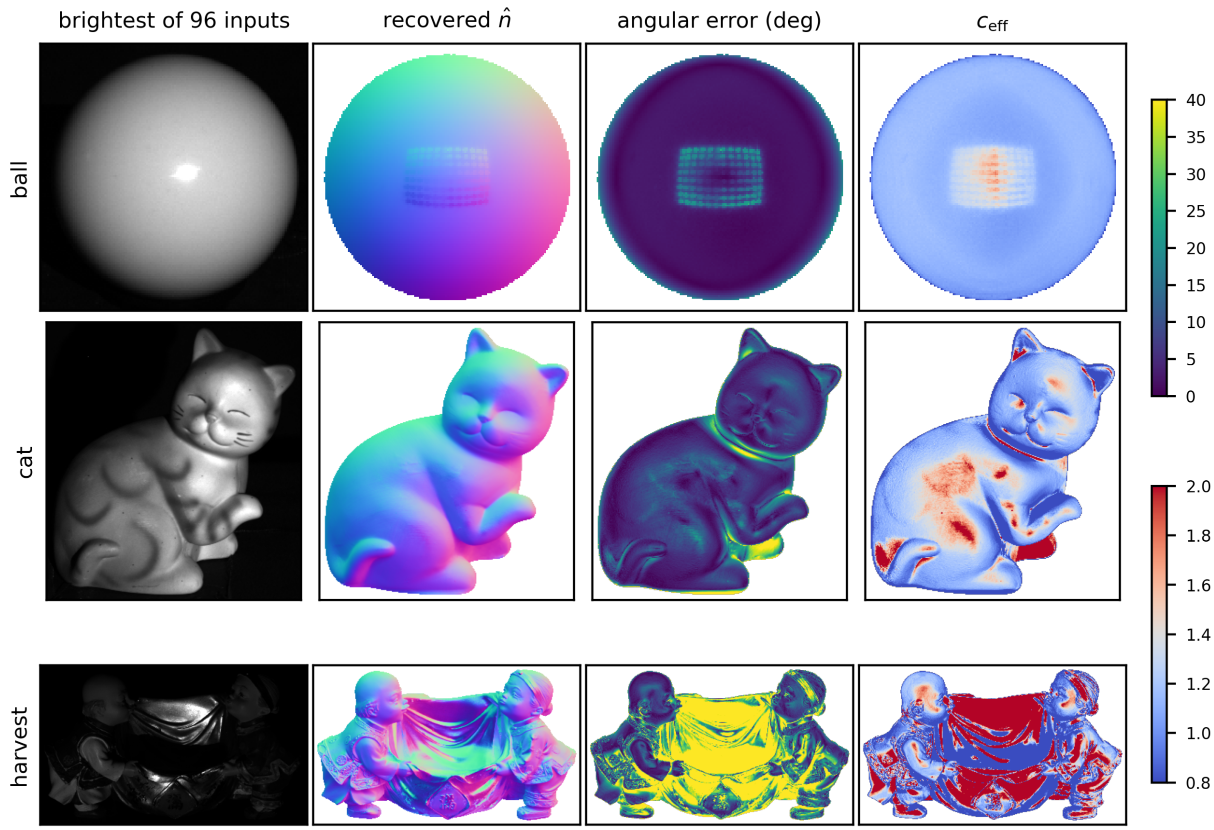

Figure 5. Three benchmark objects, spanning the reflectance range. The rightmost column is the per pixel effective exponent, and it is what a per object median cannot show. On the near diffuse ball it is nearly uniform and close to one. On the cat it spikes exactly on the glossy painted patches. On harvest the cloth saturates the scale while the two figures stay low. In every case the exponent map predicts the error map beside it, which is the spatial structure that Section 4.6 argues a single global exponent cannot capture.

The real data finding is a diagnosis. Fitted exponents above one say that the benchmark's materials sit on the far side of Lambertian from the Minnaert family, and Figure 6 says no single exponent accounts for any object. Spatial variation of the exponent accounts for most of it: an oracle per pixel exponent would outperform every classical method rescored here. That gap between 9.02 and 18.30 degrees, between the bound and the naive estimator of the same quantity, is in our view the most useful number in the paper: it locates the open problem precisely in robust exponent estimation, not in the correction itself, and the failure mode is instructive, a feedback loop in which the exponent inherits the bias of the normals it was fitted against. Regularized or trim seeded estimation is the natural next step, and DiLiGenT102 [10], with its factorial design over shapes and materials, would separate the material effect from the geometry effect that our tilt statistics only begin to control.

For a practitioner the measurements suggest a short triage, in increasing order of effort. Check conditioning first, since it is free and an ill conditioned rig dominates reflectance entirely. Then fit the exponent map: it will not yield a usable correction, but its spatial variation is diagnostic on its own, uniform for a material the linear model suits and structured for one it does not. If it is structured, trimmed least squares costs one line and recovers rimAvg degrees against 15.37 for the plain solve, and the remainder of the gap to 9.02 needs the exponent field itself.

Reproducibility deserves a note, because several results depend on it. Every quantitative result in this paper is emitted by the experiment code into a single machine readable record and typeset from that record, so a rerun cannot leave a stale figure in the tables or the prose. One command regenerates every figure, table and number from scratch. Each claim the text makes is additionally encoded as an executable assertion with a stated expectation, 101 in total, covering the exact recovery at $c = 1$, the closed form collapse at $c = 0$, the invariance and albedo predictions of Equation (6), the fitted ablation exponents, the crossover in shadow handling, exponent recovery on synthetic data, and every cell of Table 2. This is what made the two proto-

Object	c_{eff}	LS	Oracle	Trimmed
ball	1.12	4.10	2.96	2.08
bear	1.08	8.39	5.17	6.51
buddha	1.15	14.92	10.04	11.54
cat	1.18	8.41	5.06	6.90
cow	2.14	25.60	9.62	20.83
goblet	1.23	18.50	9.10	14.50
harvest	1.21	30.62	21.96	25.49
pot1	1.14	8.89	5.65	7.62
pot2	1.24	14.49	6.44	11.02
reading	1.17	19.80	14.19	13.76
Average		15.37	9.02	12.03

Table 3. Effective exponent and the correction study on the real objects, mean angular error in degrees. LS is the unmodified baseline. Oracle raises each observation to $1/c_{\text{eff}}(x)$ using the per pixel fit, which reads the ground truth and therefore bounds, rather than constitutes, a method. Trimmed drops the darkest and brightest quarter of observations per pixel and uses no reflectance model at all.

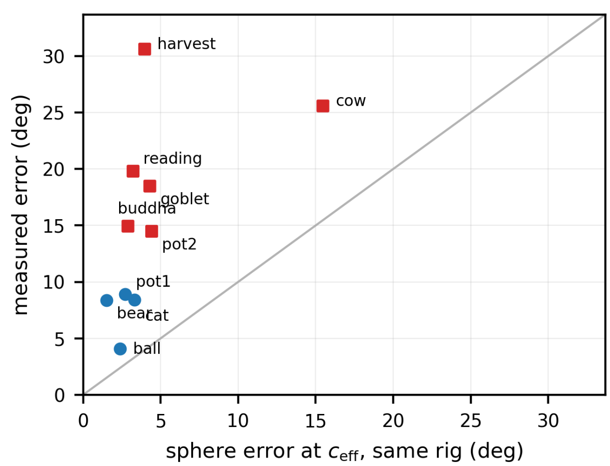

Figure 6. Does one exponent explain a real object? For each object the sphere is rendered at that object's median c_{eff} under that object's own 96 light rig, and the resulting error (horizontal) is compared with the measured one (vertical). Every object sits above the diagonal: a single global exponent underexplains the measured error by $1.7\times$ to $7.7\times$, least for the diffuse objects (circles) and most for the specular ones (squares).

col findings of Section 4.3 visible at all: both surfaced as disagreements against the reference maps at a tolerance no summary statistic would have exposed, and the degenerate pixel count is now an assertion rather than an observation.

Limitations. Everything here assumes a fixed viewpoint, distant calibrated lights, and attached shadows only; interreflections and cast shadows are absent from the synthetic data and unmodeled on the real data. The reflectance family is the simplified Hapke and Minnaert line, one exponent, no specular lobe, and the first order law is accurate only near $c = 1$, understating the error by $1.71\times$ at the far end of the range. The oracle correction is a bound, not a method, and the trim fractions of the robust control are conventional rather than tuned. Error magnitudes on any geometry depend on the range of orientations it presents, which we control with tilt statistics but do not eliminate.

6. Conclusion

We asked how much the Lambertian assumption costs photometric stereo, and made the question answerable by reducing a classical reflectance model to a single scalar acting on the solver's input. The answer has a clean synthetic half, a curve with provable endpoints and a first order law with a stated range, and a messier real half, materials that leave the model's domain, a per pixel structure no global parameter captures, and an oracle bound showing that the one parameter family, estimated well, would explain more than a decade of classical corrections. The distance between that bound and the failed naive estimator is a precisely located open problem, which is what a diagnostic study should leave behind.

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